

Product Specification: High-Frequency Real-Time Live Auction Platform

1. Executive Summary

This document specifies the features, user personas, operational flows, data entities, and technical systems for a state-of-the-art, high-frequency, real-time live auction platform. The platform enables sellers to stream high-fidelity live video/audio rooms to prospective bidders, accept ultra-low latency bids, settle transaction transactions via integrated digital wallets, enforce security through a KYC framework, and leverage an AI-powered co-pilot assistant for market analytics and automated bidding recommendations.

2. Platform Core Purpose & Objectives

- Low Latency & Engagement:** Minimize streaming and bidding latency to sub-second levels to support live virtual auction rooms.
- Financial Security:** Restrict bidding to users with verified KYC and sufficient digital wallet balances, holding bid amounts in escrow during active sessions to eliminate default risks.
- AI Copilot Co-Presence:** Assist buyers and sellers with intelligent price suggestions, predictive bidding analytics, and semantic discovery of catalog items.
- Regulatory & Compliance Integrity:** Maintain strict audit logs of all wallet credits, debits, holds, KYC submissions, and bid histories.

3. User Personas & Roles

Role	Core Purpose	Permissions & Key Actions
Bidder (Buyer)	Search, watch streams, deposit money, place bids, and buy subscription plans.	• Submit KYC documents. • Deposit/debit digital wallet via Razorpay. • Place real-time bids. • Subscribe to membership tiers.
Seller	Create auctions, start live video streams, and manage listings.	• Create auction listings with starting prices and increments. • Stream live video/audio to room participants. • Accept, reject, or mark auctions as sold. • Monitor bidding analytics.

Admin / Moderator	Manage platform configuration, approve KYC, resolve disputes.	• Review and approve/reject KYC submissions. • Adjust system parameters (minimum bid increments, fees). • Moderate fraud reports and flag items or users.
AI Assistant (Bot)	Provide semantic search, recommendations, and analytics.	• Read catalog data and historical bid listings. • Process user search intents vectorially. • Suggest optimal bid increments.

4. Product Features & Functional Requirements

4.1. Ultra-Low Latency Live Rooms (Mediasoup WebRTC)

To recreate an in-person auction house, the platform supports real-time streaming rooms.

- **Low-Latency Streaming:** Sellers broadcast video/audio directly using WebRTC via a SFU (Selective Forwarding Unit) media server running `mediasoup`.
- **Dynamic Room Lifecycle:** Sellers can initiate, pause, resume, and terminate live streams. Bidders dynamically join/leave rooms as consumers.
- **Active Bidders Panel:** Displays the live participant count, connected user profiles, and active bid leaders.

4.2. Escrow-Backed Digital Wallet System

A closed-loop wallet ecosystem guarantees all placed bids are fully collateralized.

- **Instant Wallet Deposits:** Users load funds in Indian Rupees (INR) using the secure Razorpay integration.
- **Bid Escrow Holds:** Placed bids lock the bid amount (`heldBalance`) from the bidder's `mainBalance`.
- **Automated Settle & Refund:**
 - **On Outbid:** The previously leading bidder's escrow hold is immediately released back to their `mainBalance`.
 - **On Auction Close:** The winning bidder's held balance is converted into a debit transaction and transferred to the seller's wallet, deducting platform service fees.

4.3. Interactive AI Assistant (ChromaDB + Gemini LLM)

An AI companion is embedded in the platform to assist bidders with smart features:

- **Semantic Discovery:** Bidders can issue natural language prompts (e.g., *"Find me mid-century modern furniture under ₹10,000"*) matching items vectorially using `ChromaDB` embeddings.
- **Predictive Pricing Co-pilot:** Recommends maximum bid limits and alerts users if they are overpaying based on historic transaction trends.
- **Conversational Interaction:** Real-time Q&A regarding auction details, product history, shipping estimates, and seller trust ratings.

4.4. KYC Verification Engine

Ensures a secure, fraud-free marketplace:

- **State Machine Lifecycle:** Submissions proceed through `PENDING` -> `APPROVED` or `REJECTED`.
- **Evidence Upload:** Bidders/Sellers must upload government-issued photo identification and address details.

- **Bidding Restriction:** Core bidding and selling privileges are strictly disabled until the KYC status changes to `APPROVED`.

4.5. High-Frequency Bidding Engine

- **Incremental Safeguards:** Enforces strict minimum bid increments dynamically based on current bids.
- **Concurrency Handling:** Employs optimistic/pessimistic lock mechanisms to prevent race conditions during sub-second bid placement.
- **Anti-Snipe Extensions:** If a bid is placed in the final 30 seconds of an auction, the countdown timer is automatically extended by an additional 60 seconds.

5. System Architecture & Tech Stack



6. Core Flow Charts

6.1. Bid Placement & Escrow Hold Flow



7. Data Models & Core Domain Entities

The system follows **Domain-Driven Design (DDD)** principles.

7.1. User Entity

Represents users on the platform:

- `id`: string (UUID or Nanoid)
- `email`: Email (Value Object)
- `phone`: Phone (Value Object)
- `role`: UserRole (`BIDDER`, `SELLER`, `ADMIN`)
- `kycStatus`: KycStatus (`NOT_SUBMITTED`, `PENDING`, `APPROVED`, `REJECTED`)

7.2. KYC Entity

Stores compliance materials:

- `id`: string
- `userId`: string (FK)
- `idType`: string (e.g., `AADHAAR`, `PAN`, `PASSPORT`)
- `idNumber`: string (encrypted)
- `documentUrl`: string
- `status`: KycStatus
- `rejectionReason`: string

7.3. Wallet & Transaction Entities

Monitors funds:

- **Wallet:**

- `id`: string
- `userId`: string (FK)
- `mainBalance`: number (decimal)
- `heldBalance`: number (decimal)
- `currency`: WalletCurrency (`INR`, `USD`)

- **WalletTransaction:**

- `id`: string
- `walletId`: string (FK)
- `amount`: number
- `type`: TransactionType (`DEPOSIT`, `WITHDRAWAL`, `ESCROW_HOLD`, `ESCROW_RELEASE`, `SETTLEMENT`)
- `referenceId`: string

7.4. Auction & Bid Entities

Governs bidding actions:

- **Auction:**

- `id`: string
- `sellerId`: string
- `title`: string
- `description`: string
- `startingPrice`: number
- `minIncrement`: number
- `currentBid`: number
- `status`: AuctionStatus (`DRAFT`, `ACTIVE`, `COMPLETED`, `CANCELLED`)
- `endsAt`: Date

- **Bid:**

- `id`: string
- `auctionId`: string
- `bidderId`: string
- `amount`: number
- `createdAt`: Date

8. Non-Functional Requirements

8.1. Security & Compliance

- **Data Encryption:** Encrypt all personally identifiable information (PII) such as government IDs using industry-standard symmetric encryption (`AES-256`) before database write.
- **Secure Cookies:** Deliver session tokens and credentials exclusively via `HTTPOnly`, `Secure`, and `SameSite=Lax` cookies.
- **Audit Trails:** All wallet modifications must have an associated transaction record ledger entry.